EOSC 213 – Computational Methods for Geological Engineering

Homework 02 – Solution

Problem 1: ODE Detective – Identifying Structure, Order, and Linearity

(a) Separable or not?

For each of the following ODEs, determine whether it is **separable** or **not separable**. If you claim the equation is *separable*, briefly show how it can be rearranged into the form,

$$g(y)\,dy = h(t)\,dt,$$

and state the expressions $g(y)$ and $h(t)$ in each of those cases. No need to give an explanation if it is *not separable*.

(1) $\frac{dy}{dt} = (t^2+1)\,y^3$

(2) $\frac{dy}{dt} = \sin t/(1+y^2)$

(3) $(t+1)\frac{dy}{dt} = y^2/(1+y)$

(4) $\frac{dy}{dt} = y + t^2$

(5) $\frac{dy}{dt} = t\,e^{ty}$

(6) $\frac{dy}{dt} = (y-t)/(y+t)$

(b) Order and linearity

An ODE is **linear in** y if it can be written as,

$$a_n(t)\,y^{(n)} + a_{n-1}(t)\,y^{(n-1)} + \cdots + a_1(t)\,y' + a_0(t)\,y = g(t),$$

where the coefficients $a_k(t)$ depend only on t, and y and its derivatives $y^{(k)}$ (where the superscript (k) denotes the kth derivative) appear only to the **first power** (not multiplied together and not inside nonlinear functions).

For each of the following differential equations, state (i) the **order** of the equation, and (b) whether it is **linear** or **nonlinear** in y.

(1) $y'' + t\,y' - y = e^t$

(2) $(y')^2 + y = t$

(3) $y''' + y\,y'' = 0$

(4) $\frac{d}{dt}(t\,y) + y = \sin t$

(5) $y'' + (y')^3 = 0$

(6) $t^2 y'' + t y' + y = 0$

Note: No solving is required in Problem 1. Your task is purely classification and justification.

Solution 1

(a) Separable or not?

(1) **Separable.**

$$\frac{dy}{dt} = (t^2+1)y^3 \quad \Longrightarrow \quad y^{-3}\,dy = (t^2+1)\,dt.$$

Hence $g(y) = y^{-3}$ and $h(t) = t^2 + 1$.

(2) **Separable.**

$$\frac{dy}{dt} = \frac{\sin t}{1+y^2} \quad \Longrightarrow \quad (1+y^2)\,dy = \sin t\,dt.$$

Hence $g(y) = 1 + y^2$ and $h(t) = \sin t$.

(3) **Separable.**

$$(t+1)\frac{dy}{dt} = \frac{y^2}{1+y} \quad \Longrightarrow \quad \frac{1+y}{y^2}\,dy = \frac{1}{t+1}\,dt.$$

Hence $g(y) = \dfrac{1+y}{y^2}$ and $h(t) = \dfrac{1}{t+1}$.

(4) **Not separable.**

(5) **Not separable.**

(6) **Not separable.**

(b) Order and linearity

(1) $y'' + t\,y' - y = e^t$.
Order: 2. **Linearity:** Linear (coefficients depend only on t, and y, y', y'' appear to first power).

(2) $(y')^2 + y = t$.
Order: 1. **Linearity:** Nonlinear (because $(y')^2$ is a square of a derivative).

(3) $y''' + y\,y'' = 0$.
Order: 3. **Linearity:** Nonlinear (product $y\,y''$ multiplies y and a derivative).

(4) $\dfrac{d}{dt}(t\,y) + y = \sin t.$

First expand:

$$\frac{d}{dt}(ty) = ty' + y \quad \Longrightarrow \quad ty' + y + y = \sin t \quad \Longrightarrow \quad ty' + 2y = \sin t.$$

Order: 1. **Linearity:** Linear.

(5) $y'' + (y')^3 = 0.$
Order: 2. **Linearity:** Nonlinear (because $(y')^3$ is a power of a derivative).

(6) $t^2y'' + ty' + y = 0.$
Order: 2. **Linearity:** Linear (coefficients $t^2, t, 1$ depend only on t).

Problem 2: ODE Solver - Analytical Solutions of Initial Value Problems

In this problem, you will solve a collection of ODEs *analytically*. Each equation is accompanied by an initial condition. Your task is to identify appropriate solution methods (such as separation of variables, partial fraction decomposition, or the integrating factor method), carry out the necessary algebra and integrations, and determine the solution ($y(t)$) that satisfies the given initial condition. Show all essential steps clearly and justify any transformations you use. For this problem, assume $t \geq 0$. You should clearly state any additional restrictions on t that might arise during the solution process.

(1) $\frac{dy}{dt} = \big((y+1)(y+3)\big)/\big(t(y+2)\big), \quad y(1) = 0.$

(2) $\frac{dy}{dt} = (y/t) + (y/t)^2, \quad y(1) = 1.$ Are there values of t in $t \geq 0$ where the final solution $y(t)$ is undefined?

(3) $\frac{dy}{dt} = t^4 + 3t^2 + 1 - 2ty - \frac{y}{t}, \quad y(1) = 0.$ Are there values of t in $t \geq 0$ where the final solution $y(t)$ is undefined?

(4) $\frac{dy}{dt} = t - \frac{y}{t} - y\tan t, \quad y(\frac{\pi}{6}) = 0.$ For this problem, you won't be able to compute the final integral in terms of elementary functions (like polynomials, logarithms, exponentials, or trigonometric functions), so feel free to leave the final solution of $y(t)$ in the integral form as a function of t. Are there values of t in $t \geq 0$ where the final solution $y(t)$ is undefined?

Solution 2

(1) $\dfrac{dy}{dt} = \dfrac{(y+1)(y+3)}{t(y+2)}, \quad y(1) = 0.$

Step 1: Separate variables.

$$\frac{dy}{dt} = \frac{(y+1)(y+3)}{t(y+2)} \quad \Longrightarrow \quad \frac{y+2}{(y+1)(y+3)}\,dy = \frac{1}{t}\,dt.$$

Step 2: Partial fraction decomposition (in y). We write

$$\frac{y+2}{(y+1)(y+3)} = \frac{A}{y+1} + \frac{B}{y+3}.$$

Multiply both sides by $(y+1)(y+3)$:

$$y+2 = A(y+3) + B(y+1) = (A+B)y + (3A+B).$$

Match coefficients of y and constants:

$$A + B = 1, \qquad 3A + B = 2.$$

Subtract the first equation from the second:

$$(3A+B) - (A+B) = 2 - 1 \implies 2A = 1 \implies A = \frac{1}{2},$$

and then $B = 1 - A = \frac{1}{2}$. Hence

$$\frac{y+2}{(y+1)(y+3)} = \frac{1}{2}\frac{1}{y+1} + \frac{1}{2}\frac{1}{y+3}.$$

Step 3: Integrate both sides.

$$\int \left(\frac{1}{2}\frac{1}{y+1} + \frac{1}{2}\frac{1}{y+3}\right) dy = \int \frac{1}{t}\,dt.$$

$$\frac{1}{2}\ln|y+1| + \frac{1}{2}\ln|y+3| = \ln|t| + C.$$

Step 4: Combine logarithms and remove logs.

$$\frac{1}{2}\ln|y+1| + \frac{1}{2}\ln|y+3| = \frac{1}{2}\ln\big(|(y+1)(y+3)|\big) = \ln|t| + C.$$

Multiply by 2:

$$\ln\big(|(y+1)(y+3)|\big) = 2\ln|t| + C' \quad \Longrightarrow \quad |(y+1)(y+3)| = K\,t^2.$$

Absorb the absolute-value sign into the constant K (allowing K to be any nonzero real constant):

$$(y+1)(y+3) = Kt^2.$$

Step 5: Apply the initial condition $y(1) = 0$.

$$(0+1)(0+3) = K(1)^2 \implies 3 = K.$$

So the implicit solution is

$$(y+1)(y+3) = 3t^2.$$

Step 6: Expand and obtain a quadratic equation in y.

$$(y+1)(y+3) = y^2 + 4y + 3.$$

Thus

$$y^2 + 4y + 3 = 3t^2 \quad \implies \quad y^2 + 4y + (3 - 3t^2) = 0.$$

This is a **quadratic equation** in y.

Step 7: Solve the quadratic using the quadratic formula. For $ay^2 + by + c = 0$ with $a = 1$, $b = 4$, $c = 3 - 3t^2$,

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-4 \pm \sqrt{16 - 4(1)(3 - 3t^2)}}{2}.$$

Simplify the discriminant:

$$16 - 4(3 - 3t^2) = 16 - 12 + 12t^2 = 4 + 12t^2 = 4(1 + 3t^2).$$

Hence

$$y = \frac{-4 \pm \sqrt{4(1+3t^2)}}{2} = \frac{-4 \pm 2\sqrt{1+3t^2}}{2} = -2 \pm \sqrt{1+3t^2}.$$

Step 8: Choose the correct branch using $y(1) = 0$.

$$y(1) = -2 \pm \sqrt{1 + 3(1)^2} = -2 \pm 2.$$

To get $y(1) = 0$, we must choose the $+$ sign. Therefore

$$\boxed{y(t) = -2 + \sqrt{1+3t^2}}.$$

Step 9: Domain restriction (for $y(t)$, with $t \geq 0$): Since $1 + 3t^2 > 0$ for all $t \geq 0$, the square root is always defined. Thus

$$\boxed{y(t) \text{ is defined for all } t \geq 0.}$$

For part (1), discussing domain restriction on $y(t)$ is not required to obtain full marks since it was not mentioned in the question.

(2) $\dfrac{dy}{dt} = \dfrac{y}{t} + \left(\dfrac{y}{t}\right)^2, \quad y(1) = 1.$

Step 1: Introduce a substitution to simplify the expression $\frac{y}{t}$.

Because the right-hand side depends only on $\frac{y}{t}$, let

$$v(t) = \frac{y(t)}{t} \quad \Longleftrightarrow \quad y(t) = t\,v(t).$$

Differentiate $y = tv$ using the product rule:

$$\frac{dy}{dt} = \frac{d}{dt}(tv) = v + t\frac{dv}{dt}.$$

Step 2: Substitute into the ODE and simplify.

The ODE becomes

$$v + t\frac{dv}{dt} = v + v^2.$$

Subtract v from both sides:

$$t\frac{dv}{dt} = v^2.$$

Step 3: Separate variables.

Assuming $v \neq 0$ (we will get the general solution anyway), divide by v^2 and multiply by dt:

$$\frac{1}{v^2}\,dv = \frac{1}{t}\,dt.$$

Step 4: Integrate both sides.

$$\int v^{-2}\,dv = \int \frac{1}{t}\,dt.$$

Compute each integral:

$$\int v^{-2}\,dv = -v^{-1} = -\frac{1}{v}, \qquad \int \frac{1}{t}\,dt = \ln|t| + C.$$

So

$$-\frac{1}{v} = \ln|t| + C.$$

Step 5: Solve for $v(t)$.

Multiply by -1:

$$\frac{1}{v} = -(\ln|t| + C).$$

Invert both sides:

$$v(t) = \frac{1}{-(\ln|t| + C)} = -\frac{1}{\ln|t| + C}.$$

Step 6: Convert back to $y(t)$.

Since $y = tv$,

$$y(t) = t\left(-\frac{1}{\ln|t| + C}\right) = -\frac{t}{\ln|t| + C}.$$

Step 7: Use the initial condition $y(1) = 1$ to determine C.

Plug in $t = 1$ and $y(1) = 1$:

$$1 = -\frac{1}{\ln|1| + C}.$$

Since $\ln|1| = 0$,

$$1 = -\frac{1}{C} \quad \Longrightarrow \quad C = -1.$$

Therefore,

$$\boxed{y(t) = -\frac{t}{\ln|t| - 1}}.$$

Step 8: Domain restriction (for $y(t)$, with $t \geq 0$).

Because $t \geq 0$, we have $|t| = t$. The expression

$$y(t) = -\frac{t}{\ln t - 1}$$

is undefined when:

- $\ln t$ is undefined $\Rightarrow t = 0$ is not allowed (we need $t > 0$).
- the denominator is zero:

$$\ln t - 1 = 0 \quad \Longrightarrow \quad \ln t = 1 \quad \Longrightarrow \quad t = e.$$

Hence, within $t \geq 0$,

$$\boxed{y(t) \text{ is undefined at } t = 0 \text{ and } t = e.}$$

(3) $\dfrac{dy}{dt} = t^4 + 3t^2 + 1 - 2ty - \dfrac{y}{t}, \quad y(1) = 0.$

Step 1: Rewrite in standard linear form.

Bring the y-terms to the left:

$$\frac{dy}{dt} + 2t\,y + \frac{1}{t}y = t^4 + 3t^2 + 1.$$

So the ODE is

$$y' + \left(2t + \frac{1}{t}\right)y = t^4 + 3t^2 + 1,$$

which is a first-order **linear** ODE of the form $y' + p(t)y = q(t)$ with

$$p(t) = 2t + \frac{1}{t}, \qquad q(t) = t^4 + 3t^2 + 1.$$

Step 2: Compute the integrating factor $\mu(t)$.

$$\mu(t) = \exp\left(\int p(t)\,dt\right) = \exp\left(\int \left(2t + \frac{1}{t}\right) dt\right).$$

Compute the integral:

$$\int \left(2t + \frac{1}{t}\right) dt = t^2 + \ln|t|.$$

Therefore

$$\mu(t) = e^{t^2 + \ln|t|} = e^{t^2}\, e^{\ln|t|} = |t| e^{t^2}.$$

Since the problem assumes $t \geq 0$, we have $|t| = t$, so

$$\mu(t) = te^{t^2}.$$

Step 3: Multiply the ODE by $\mu(t)$ and use the product-derivative identity.

Multiply both sides of

$$y' + p(t)y = q(t)$$

by $\mu(t)$:

$$\mu y' + \mu p(t)\, y = \mu q(t).$$

By construction of μ, the left-hand side becomes

$$\frac{d}{dt}\big(\mu(t)y(t)\big).$$

Hence

$$\frac{d}{dt}\big(te^{t^2} y\big) = te^{t^2}(t^4 + 3t^2 + 1).$$

Step 4: Integrate both sides.

$$te^{t^2} y = \int te^{t^2}(t^4 + 3t^2 + 1)\,dt + C.$$

Step 5: Evaluate the integral using the substitution $u = t^2$.

Let

$$u = t^2 \quad \Longrightarrow \quad du = 2t\,dt \quad \Longrightarrow \quad t\,dt = \frac{1}{2}\,du.$$

Also note:

$$t^4 = (t^2)^2 = u^2, \qquad 3t^2 = 3u.$$

Therefore,

$$\int te^{t^2}(t^4 + 3t^2 + 1)\,dt = \frac{1}{2}\int e^u(u^2 + 3u + 1)\,du.$$

Now compute $\int e^u(u^2 + 3u + 1)\,du$ using **integration by parts**.

First split the integral:

$$\int e^u(u^2+3u+1)\,du = \int u^2e^u\,du + 3\int ue^u\,du + \int e^u\,du.$$

1) Compute $\int u^2e^u\,du$ by parts (twice).

Let $A = \int u^2e^u\,du$. Take $f = u^2$, $dg = e^u\,du$, so $df = 2u\,du$, $g = e^u$. Then

$$A = u^2e^u - \int 2ue^u\,du.$$

Now let $B = \int ue^u\,du$. Again integrate by parts with $f = u$, $dg = e^u\,du$:

$$B = ue^u - \int e^u\,du = ue^u - e^u.$$

Substitute B back into A:

$$A = u^2e^u - 2B = u^2e^u - 2(ue^u - e^u) = e^u(u^2 - 2u + 2).$$

2) Compute $\int ue^u\,du$ (already found).

$$\int ue^u\,du = ue^u - e^u = e^u(u-1).$$

3) Compute $\int e^u\,du$.

$$\int e^u\,du = e^u.$$

4) Combine the pieces.

$$\int e^u(u^2+3u+1)\,du = e^u(u^2-2u+2) + 3e^u(u-1) + e^u.$$

Factor out e^u and simplify inside:

$$= e^u\Big[(u^2-2u+2) + (3u-3) + 1\Big] = e^u(u^2+u).$$

Therefore,

$$\int te^{t^2}(t^4+3t^2+1)\,dt = \frac{1}{2}e^u(u^2+u) + C = \frac{1}{2}e^{t^2}(t^4+t^2) + C.$$

Step 6: Substitute back and solve for $y(t)$.

From Step 4:

$$te^{t^2}y = \frac{1}{2}e^{t^2}(t^4+t^2) + C.$$

Divide both sides by te^{t^2}:

$$y(t) = \frac{\frac{1}{2}e^{t^2}(t^4+t^2)}{te^{t^2}} + \frac{C}{te^{t^2}} = \frac{1}{2}\frac{t^4+t^2}{t} + \frac{C}{t}e^{-t^2}.$$

Simplify:

$$y(t) = \frac{1}{2}(t^3+t) + \frac{C}{t}e^{-t^2}.$$

Step 7: Use the initial condition $y(1) = 0$ to determine C.

Plug in $t = 1$:

$$0 = \frac{1}{2}(1^3+1) + \frac{C}{1}e^{-1} = \frac{1}{2}(2) + Ce^{-1} = 1 + Ce^{-1}.$$

So

$$C = -e.$$

Therefore,

$$\boxed{y(t) = \frac{1}{2}(t^3+t) - \frac{e^{1-t^2}}{t}}.$$

Step 8: Domain restriction (for $y(t)$, with $t \geq 0$).

The expression contains a factor $1/t$, so it is undefined at $t = 0$. For all $t > 0$, the exponential and polynomial terms are defined.

$$\boxed{y(t) \text{ is undefined at } t = 0 \text{ (and defined for all } t > 0).}$$

(4) $\dfrac{dy}{dt} = t - \dfrac{y}{t} - y\tan t, \quad y\left(\dfrac{\pi}{6}\right) = 0.$

Step 1: Rewrite the ODE in standard linear form.

Start with

$$\frac{dy}{dt} = t - \frac{y}{t} - y\tan t.$$

Move the y-terms to the left:

$$y' + \left(\frac{1}{t} + \tan t\right)y = t.$$

This is a first-order **linear** ODE of the form

$$y' + p(t)\,y = q(t), \qquad p(t) = \frac{1}{t} + \tan t, \qquad q(t) = t.$$

Step 2: Compute the integrating factor $\mu(t)$.

$$\mu(t) = \exp\left(\int p(t)\,dt\right) = \exp\left(\int \left(\frac{1}{t} + \tan t\right)dt\right).$$

Compute the integrals:

$$\int \frac{1}{t}\,dt = \ln|t|, \qquad \int \tan t\,dt = \int \frac{\sin t}{\cos t}\,dt = -\ln|\cos t|.$$

Therefore,

$$\int p(t)\,dt = \ln|t| - \ln|\cos t| = \ln\left(\frac{|t|}{|\cos t|}\right).$$

Exponentiate:

$$\mu(t) = \exp\left(\ln\left(\frac{|t|}{|\cos t|}\right)\right) = \frac{|t|}{|\cos t|}.$$

Since the problem assumes $t \geq 0$, we have $|t| = t$, so we may write

$$\mu(t) = \frac{t}{|\cos t|}.$$

On any interval where $\cos t \neq 0$ and does not change sign, we can drop the absolute value consistently. In particular, near $t = \pi/6$ we have $\cos t > 0$, so we take

$$\mu(t) = \frac{t}{\cos t} = t\sec t \qquad \text{(on intervals where } \cos t > 0\text{)}.$$

Step 3: Multiply the ODE by $\mu(t)$ and convert the left side to a product derivative.

Multiply

$$y' + p(t)y = q(t)$$

by $\mu(t)$:

$$\mu y' + \mu p(t)\,y = \mu q(t).$$

By the integrating factor construction,

$$\mu y' + \mu p(t)\,y = \frac{d}{dt}\big(\mu y\big).$$

So we obtain

$$\frac{d}{dt}\big(t\sec t\,y(t)\big) = t\sec t \cdot t = t^2\sec t.$$

Step 4: Integrate both sides and use the initial condition.

Integrate from the initial time $t_0 = \pi/6$ to t:

$$\int_{t_0}^{t} \frac{d}{ds}\big(s\sec s\,y(s)\big)\,ds = \int_{t_0}^{t} s^2\sec s\,ds.$$

By the Fundamental Theorem of Calculus,

$$t\sec t\,y(t) - t_0\sec t_0\,y(t_0) = \int_{\pi/6}^{t} s^2\sec s\,ds.$$

Since $y(\pi/6) = 0$, the second term vanishes:

$$t \sec t \, y(t) = \int_{\pi/6}^{t} s^2 \sec s \, ds.$$

Step 5: Solve for $y(t)$.

Divide by $t \sec t$:

$$y(t) = \frac{1}{t \sec t} \int_{\pi/6}^{t} s^2 \sec s \, ds = \frac{\cos t}{t} \int_{\pi/6}^{t} s^2 \sec s \, ds.$$

Thus,

$$\boxed{y(t) = \frac{\cos t}{t} \int_{\pi/6}^{t} s^2 \sec s \, ds}.$$

Step 6: Domain restriction (for $y(t)$, with $t \geq 0$).

From the final expression

$$y(t) = \frac{\cos t}{t} \int_{\pi/6}^{t} s^2 \sec s \, ds,$$

$y(t)$ is undefined when:

- $t = 0$ (because of the factor $1/t$),
- the integral is not defined, which occurs when $\sec s$ blows up inside the integration interval, i.e. when $\cos s = 0$ for some s between $\pi/6$ and t.

The zeros of $\cos s$ are at

$$s = \frac{\pi}{2} + k\pi, \qquad k = 0, 1, 2, \ldots$$

Hence, for $t \geq 0$,

$$\boxed{y(t) \text{ is undefined at } t = 0 \text{ and at } t = \frac{\pi}{2} + k\pi, \ k = 0, 1, 2, \ldots}.$$

Problem 3: Forward Euler with Polynomially-Growing Decay

Consider the initial value problem,

$$\frac{dy}{dt} = -t^2 y, \qquad y(0) = y_0,$$

whose exact solution is,

$$y(t) = y_0 e^{-t^3/3}.$$

This is similar to the exponential decay model discussed in class but with $\lambda(t) = t^2$ (an increasing function of t; not a constant). Let $h > 0$ be a fixed time step and define the time grid $t_n = nh$, where $n = 0, 1, \ldots, N-1$. Forward Euler algorithm attempts to produce approximations y_n at time

points t_n which, depending on the accuracy of the algorithm, may or may not be approximately equal to the exact solution $y(t_n)$ at time points t_n.

(a) Forward Euler iteration

Derive the Forward Euler update formula and express y_n explicitly in terms of a product.

(b) Loss of monotone decay

Show that for any fixed $h > 0$, there exists an index k such that the Euler factor

$$1 - h(kh)^2$$

becomes negative. Explain the consequence for the numerical solution and contrast it with the exact solution.

(c) Exact step-to-step amplification

Compute the exact amplification factor,

$$\frac{y(t_{n+1})}{y(t_n)}$$

and compare it with the Forward Euler amplification factor at step n. Comment on how they differ as n increases.

(d) Local truncation error (single-step error)

Assume the exact value $y(t_n)$ is known. The *local truncation error* incurred on the step from t_n to t_{n+1} is defined as the difference between the exact solution and a single Forward Euler step starting from the exact value at t_n,

$$\tau_{n+1} := y(t_{n+1}) - \underbrace{\big(y(t_n) + h\, f(t_n, y(t_n))\big)}_{y(t_{n+1}) \text{ computed via the Forward Euler step}}$$

(where $f(t, y) = -t^2 y$ for this problem). Compute τ_{n+1} explicitly and also determine its leading-order behavior in terms of h and n. You may use the expansion $\exp(-x) = 1 - x + \frac{x^2}{2} + \mathcal{O}(x^3)$.

(e) Error propagation (global error)

Define the *global error* at time t_n as,

$$e_n := y(t_n) - y_n,$$

that is, the difference between the exact solution and the numerical approximation at the same time step. Derive a recursion for e_{n+1} in terms of e_n and τ_{n+1}, and explain why numerical errors

may grow even though the exact solution is decaying.

Remark (local vs. global error): *The local truncation error measures the error introduced by a* single time step *of the numerical method, assuming the exact solution is known at the previous time step. In contrast, the global error measures the difference between the numerical solution and the exact solution at a given time, after errors from all previous steps have accumulated.*

(f) Interpretation

Briefly explain what this example reveals about the limitations of the Forward Euler method for long-time integration of stable ODEs with time-dependent coefficients.

Solution 3

We consider,

$$y'(t) = -t^2 y(t), \qquad y(0) = y_0, \qquad t_n = nh.$$

The exact solution is,

$$y(t) = y_0 e^{-t^3/3}.$$

(a) Forward Euler iteration

Forward Euler uses,

$$y_{k+1} = y_k + hf(t_k, y_k), \qquad f(t, y) = -t^2 y.$$

Hence,

$$y_{k+1} = y_k - ht_k^2 y_k = (1 - ht_k^2)\, y_k.$$

Since $t_k = kh$, this can be written as,

$$y_{k+1} = (1 - h(kh)^2)\, y_n = (1 - h^3 k^2)\, y_k.$$

Iterating from $k = 0$ to $n - 1$ gives the product form,

$$\boxed{y_n = y_0 \prod_{k=0}^{n-1} \left(1 - ht_k^2\right) = y_0 \prod_{k=0}^{n-1} \left(1 - h^3 k^2\right).}$$

There is no single constant amplification factor; instead, the step-to-step factor depends on n.

(b) Loss of monotone decay

The Forward Euler factor at step k is,

$$1 - ht_k^2 = 1 - h(kh)^2 = 1 - h^3 k^2.$$

This becomes negative whenever,

$$1 - h^3k^2 < 0 \quad \iff \quad h^3k^2 > 1 \quad \iff \quad k > \frac{1}{h^{3/2}}.$$

Thus, for any fixed $h > 0$, there exists an integer k large enough (e.g. $k = \lceil h^{-3/2} \rceil + 1$) such that,

$$1 - h^3k^2 < 0.$$

Consequences:

- Once a factor becomes negative, the numerical solution y_n may change sign (oscillate), so it can lose monotone decay.
- In contrast, the exact solution $y(t) = y_0 e^{-t^3/3}$ is strictly positive for all $t \geq 0$ if $y_0 > 0$, and strictly negative for all $t \geq 0$ if $y_0 < 0$; it never changes sign and decays monotonically in magnitude.

(c) Exact step-to-step amplification

The exact amplification from t_n to t_{n+1} is,

$$\frac{y(t_{n+1})}{y(t_n)} = \frac{y_0 e^{-t_{n+1}^3/3}}{y_0 e^{-t_n^3/3}} = \exp\left(-\frac{t_{n+1}^3 - t_n^3}{3}\right).$$

Now $t_{n+1} = (n+1)h$ and $t_n = nh$, so,

$$t_{n+1}^3 - t_n^3 = \big((n+1)h\big)^3 - (nh)^3 = (n+1)^3h^3 - n^3h^3 = (3n^2 + 3n + 1)h^3.$$

Therefore,

$$\boxed{A_{\text{exact}}(n) := \frac{y(t_{n+1})}{y(t_n)} = \exp\big(-h^3\left(n^2 + n + \tfrac{1}{3}\right)\big)\,.}$$

The Forward Euler amplification factor at step n is the multiplier in the update,

$$\boxed{A_{\text{FE}}(n) := 1 - ht_n^2 = 1 - h^3n^2.}$$

Comparison as n increases:

- $A_{\text{exact}}(n)$ is always positive and strictly less than 1 for $n \geq 0$, so it always produces decay without sign changes.
- $A_{\text{FE}}(n)$ eventually becomes negative (and its magnitude can exceed 1), causing sign flips and potential growth in magnitude.

(d) Local truncation error (single-step error)

By definition,

$$\tau_{n+1} := y(t_{n+1}) - \Big(y(t_n) + hf(t_n, y(t_n))\Big), \qquad f(t,y) = -t^2y.$$

Compute the Forward Euler one-step prediction starting from the exact value,

$$y(t_n) + hf(t_n, y(t_n)) = y(t_n) - ht_n^2 y(t_n) = (1 - ht_n^2)\, y(t_n).$$

Also,

$$y(t_{n+1}) = y(t_n)\, A_{\text{exact}}(n) = y(t_n) \exp\left(-h^3 \left(n^2 + n + \tfrac{1}{3}\right)\right).$$

Hence,

$$\boxed{\tau_{n+1} = y(t_n) \left[\exp\left(-h^3 \left(n^2 + n + \tfrac{1}{3}\right)\right) - (1 - h^3 n^2)\right].}$$

To get the leading-order behavior, let,

$$x := h^3 \left(n^2 + n + \tfrac{1}{3}\right).$$

Using $\exp(-x) = 1 - x + \frac{x^2}{2} + \mathcal{O}(x^3)$,

$$\begin{aligned}
\exp(-x) - (1 - h^3 n^2) &= \left(1 - x + \frac{x^2}{2} + \mathcal{O}(x^3)\right) - 1 + h^3 n^2 \\
&= -x + h^3 n^2 + \frac{x^2}{2} + \mathcal{O}(x^3) \\
&= -h^3 \left(n + \tfrac{1}{3}\right) + \frac{x^2}{2} + \mathcal{O}(x^3).
\end{aligned}$$

Therefore,

$$\boxed{\tau_{n+1} = -h^3 \left(n + \tfrac{1}{3}\right) y(t_n) + \mathcal{O}\left(h^6 n^4\right) y(t_n).}$$

In particular, the leading term scales like,

$$\tau_{n+1} = \mathcal{O}(h^3 n)\, y(t_n) \quad \text{(for large } n\text{, fixed small } h\text{)}.$$

(e) Error propagation (global error)

Define the global error by the convention "exact minus numerical",

$$e_n := y(t_n) - y_n.$$

Start from the numerical update,

$$y_{n+1} = (1 - ht_n^2)\, y_n.$$

Write the exact update in the form "one Euler step plus local truncation error",

$$y(t_{n+1}) = \Big(y(t_n) + hf(t_n, y(t_n))\Big) + \tau_{n+1} = (1 - ht_n^2)\, y(t_n) + \tau_{n+1}.$$

Now subtract the numerical value from the exact value,

$$\begin{aligned} e_{n+1} &= y(t_{n+1}) - y_{n+1} \\ &= \Big((1 - ht_n^2)y(t_n) + \tau_{n+1}\Big) - (1 - ht_n^2)y_n \\ &= (1 - ht_n^2)\big(y(t_n) - y_n\big) + \tau_{n+1} \\ &= (1 - ht_n^2)e_n + \tau_{n+1}. \end{aligned}$$

Thus the global error recursion is,

$$\boxed{e_{n+1} = (1 - ht_n^2)\, e_n + \tau_{n+1} = (1 - h^3 n^2)\, e_n + \tau_{n+1}.}$$

To see how errors accumulate, iterate this recursion,

$$\boxed{e_n = \sum_{j=0}^{n-1} \tau_{j+1} \prod_{k=j+1}^{n-1} (1 - ht_k^2),}$$

assuming $e_0 = 0$ since $y_0 = y(t_0)$ is known exactly. Explanation of possible growth include:

- The local truncation error τ_{j+1} is injected at every step.
- The multipliers $(1 - ht_k^2) = (1 - h^3 k^2)$ can eventually have magnitude ≥ 1 and can become negative.
- Therefore, previously accumulated errors can be amplified (and can oscillate in sign), even though the exact solution decays.

(f) Interpretation

Even though the continuous system is stable and decays rapidly,

$$y(t) = y_0 e^{-t^3/3} \to 0 \quad \text{as } t \to \infty,$$

Forward Euler uses time-dependent step multipliers $1 - ht_n^2$ that:

- are not uniformly bounded in magnitude by 1 for all n,
- eventually become negative for large enough n for any fixed $h > 0$,
- can amplify global errors through the recursion $e_{n+1} = (1 - ht_n^2)e_n + \tau_{n+1}$.

Hence Forward Euler can lose monotone decay and exhibit oscillations and error growth in long-time integration of stable ODEs with time-dependent coefficients, motivating either step-size restrictions that depend on the final time or the use of implicit methods.

Problem 4: Stability Bound for the Explicit Midpoint Method

In this problem, you will derive the stability bound (step-size restriction) for the *explicit midpoint method* applied to the exponential decay model (as discussed in the lectures),

$$x'(t) = -\lambda x(t), \qquad \lambda > 0.$$

Recall the explicit midpoint method (also known as the 2-stage Runge–Kutta method or RK2),

$$x_{n+\frac{1}{2}} = x_n + \frac{h}{2} f(t_n, x_n), \qquad x_{n+1} = x_n + h\, f\left(t_n + \frac{h}{2},\, x_{n+\frac{1}{2}}\right),$$

where $f(t, x) = -\lambda x$.

(a) Derive the amplification factor

Derive the explicit midpoint update for this ODE in the form,

$$x_{n+1} = g_{\mathrm{EM}}\, x_n,$$

and show that the amplification factor is,

$$g_{\mathrm{EM}}(z) = 1 - z + \frac{z^2}{2}, \qquad \text{where } z := h\lambda.$$

(b) Solve the stability inequality

For a decaying solution, we require the method to be *stable* in the sense that,

$$|g_{\mathrm{EM}}(z)| \leq 1.$$

Solve the inequality,

$$\left|1 - z + \frac{z^2}{2}\right| \leq 1$$

explicitly for real $z \geq 0$, and hence obtain the stability bound on the step size h in terms of λ.

(c) Interpret the result

Using your result from parts (a)–(b) for the explicit midpoint method and the known stability bounds on h for Forward Euler as applied to exponential decay model (discussed in the lectures), compare the two methods in terms of their stability ranges for h and determine whether the maximum stable step size $h_{\max}$ is the same or different. Briefly discuss how their different amplification factors influence the qualitative behavior of the numerical solution within the stable regime.

Solution 4

(a) Amplification factor

Here $f(t,x) = -\lambda x$. First compute the midpoint stage,

$$x_{n+\frac{1}{2}} = x_n + \frac{h}{2}(-\lambda x_n) = \left(1 - \frac{h\lambda}{2}\right) x_n.$$

Then the full step is,

$$x_{n+1} = x_n + h\left(-\lambda x_{n+\frac{1}{2}}\right) = x_n - h\lambda\left(1 - \frac{h\lambda}{2}\right) x_n.$$

Therefore,

$$x_{n+1} = \left(1 - h\lambda + \frac{(h\lambda)^2}{2}\right) x_n.$$

Defining $z = h\lambda$, the amplification factor is,

$$\boxed{g_{\mathrm{EM}}(z) = 1 - z + \frac{z^2}{2}.}$$

(b) Solving stability inequality

We require,

$$\left|1 - z + \frac{z^2}{2}\right| \leq 1, \qquad z \geq 0.$$

Let,

$$g(z) := 1 - z + \frac{z^2}{2}.$$

Then $|g(z)| \leq 1$ is equivalent to the pair of inequalities,

$$-1 \leq g(z) \leq 1.$$

Upper bound: $g(z) \leq 1$**.**

$$1 - z + \frac{z^2}{2} \leq 1 \quad \Longleftrightarrow \quad -z + \frac{z^2}{2} \leq 0 \quad \Longleftrightarrow \quad z\left(\frac{z}{2} - 1\right) \leq 0.$$

For $z \geq 0$, this gives

$$0 \leq z \leq 2.$$

Lower bound: $g(z) \geq -1$**.**

$$1 - z + \frac{z^2}{2} \geq -1 \quad \Longleftrightarrow \quad \frac{z^2}{2} - z + 2 \geq 0.$$

Multiply by 2:

$$z^2 - 2z + 4 \geq 0.$$

Its discriminant is

$$\Delta = (-2)^2 - 4(1)(4) = 4 - 16 = -12 < 0,$$

and since the leading coefficient is positive, the quadratic is strictly positive for all real z. Thus,

$$z^2 - 2z + 4 \geq 0 \quad \text{for all } z,$$

so the lower bound imposes *no additional restriction*.

Conclusion. The stability condition for real $z \geq 0$ is therefore

$$\boxed{0 \leq z \leq 2.}$$

Since $z = h\lambda$, the step-size restriction is

$$\boxed{0 < h \leq \frac{2}{\lambda}.}$$

(c) Interpretation

Forward Euler is stable when,

$$|1 - z| \leq 1 \quad \Longleftrightarrow \quad 0 \leq z \leq 2.$$

From part (b), explicit midpoint is stable when,

$$\left|1 - z + \frac{z^2}{2}\right| \leq 1 \quad \Longleftrightarrow \quad 0 \leq z \leq 2.$$

Thus, both methods allow the same maximum stable step size,

$$\boxed{h_{\max} = \frac{2}{\lambda}.}$$

However, the amplification factors differ,

$$g_{\mathrm{FE}}(z) = 1 - z, \qquad g_{\mathrm{EM}}(z) = 1 - z + \frac{z^2}{2}.$$

For small z, $g_{\mathrm{EM}}(z)$ matches the Taylor expansion of e^{-z} to higher order than $g_{\mathrm{FE}}(z)$, so within the stable range explicit midpoint typically produces a more accurate decay per step (second-order accurate) than Forward Euler (first-order accurate), even though both share the same stability bound $0 \leq h\lambda \leq 2$.

Problem 5: Backward Euler - Analytical Updates and Stability

In this problem, you will analyze the *Backward Euler* method applied to initial value problems (IVPs). Unlike Forward Euler, Backward Euler evaluates the right-hand side at the future time step,

$$y_{n+1} = y_n + h\, f(t_{n+1}, y_{n+1}).$$

For the ODEs in this problem, the Backward Euler update can be solved *exactly* for y_{n+1} using algebra (no iterative solvers such as Newton's method are needed).

(a) Variable coefficient linear decay

Consider,

$$y'(t) = -t\, y(t), \qquad y(0) = y_0.$$

1. Derive the Backward Euler update formula for y_{n+1} in terms of y_n, h, and t_{n+1}.
2. Write y_n as a product involving the time grid points t_k.
3. Briefly comment on whether Backward Euler can change the sign of the solution when $y_0 > 0$.

(b) A nonlinear ODE with an explicit Backward Euler update

Consider the nonlinear IVP,

$$y'(t) = -y(t)^2, \qquad y(0) = y_0 > 0.$$

1. Derive the Backward Euler update formula for y_{n+1} in terms of y_n, h, and t_{n+1}.
2. Show that this equation is a quadratic in y_{n+1} and solve it explicitly for y_{n+1} in terms of y_n and h.
3. Select the physically meaningful branch (the one that keeps $y_{n+1} > 0$) of the quadratic equation and explain your choice briefly.

Solution 5

(a) Variable coefficient linear decay

For the model $y'(t) = -t\, y(t)$, Backward Euler gives,

$$y_{n+1} = y_n + h\big(-t_{n+1} y_{n+1}\big) \quad \Longrightarrow \quad (1 + h t_{n+1}) y_{n+1} = y_n.$$

So,

$$\boxed{y_{n+1} = \frac{1}{1 + h t_{n+1}}\, y_n.}$$

Iterating,

$$\boxed{y_n = y_0 \prod_{k=1}^{n} \frac{1}{1+ht_k} = y_0 \prod_{k=1}^{n} \frac{1}{1+kh^2}.}$$

If $y_0 > 0$, then each factor $\frac{1}{1+ht_k} > 0$, so $y_n > 0$ for all n. Thus, Backward Euler does not introduce sign changes for this decay problem.

(b) A nonlinear ODE with an explicit Backward Euler update

For the model $y'(t) = -y(t)^2$, Backward Euler gives,

$$y_{n+1} = y_n + h\big(-y_{n+1}^2\big) \quad \Longrightarrow \quad hy_{n+1}^2 + y_{n+1} - y_n = 0.$$

This is a quadratic in y_{n+1}. Solving,

$$y_{n+1} = \frac{-1 \pm \sqrt{1+4hy_n}}{2h}.$$

Since $h > 0$ and $y_n > 0$, we have $\sqrt{1+4hy_n} > 1$, so the "plus" branch yields,

$$\frac{-1+\sqrt{1+4hy_n}}{2h} > 0,$$

while the "minus" branch is negative. The physically meaningful choice (preserving positivity) is,

$$\boxed{y_{n+1} = \frac{-1+\sqrt{1+4hy_n}}{2h}.}$$

This provides an explicit closed-form Backward Euler update without any iterative solver.

Problem 6: A Linear System of ODEs – Coupling and Stability

In this problem, you will analyze a coupled system of ordinary differential equations. Such systems arise frequently in geophysics, for example in coupled reservoirs, energy balance models, or interacting chemical species. Consider the linear system,

$$\begin{cases} x'(t) = -\alpha x(t) + \beta y(t), \\ y'(t) = -\beta x(t) - \alpha y(t), \end{cases} \qquad \alpha > 0,\ \beta > 0,$$

with initial condition $(x(0), y(0)) = (x_0, y_0)$.

(a) Matrix form

Write the system in matrix form,

$$\mathbf{u}'(t) = A\,\mathbf{u}(t), \qquad \mathbf{u}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix},$$

and explicitly state the matrix A.

(b) Eigenvalues and qualitative behavior

Compute the eigenvalues of A. Based on the eigenvalues, describe the qualitative behavior of the exact solution (decay, oscillation, or both).

(c) Exact solution structure

Without computing the full closed-form solution, show that the quantity,

$$r(t)^2 := x(t)^2 + y(t)^2$$

satisfies a scalar ODE. Solve this ODE and describe how the magnitude of the solution evolves in time.
Note: A scalar ODE is an ODE involving a single unknown function of one variable (as we discussed in class until lecture 06), as opposed to a system of ODEs that may involve multiple coupled unknown functions.

(d) Forward Euler applied to the system

Apply the Forward Euler method with step size $h > 0$ to the system and write the update in matrix form,

$$\mathbf{u}_{n+1} = B_{\mathrm{FE}}\,\mathbf{u}_n.$$

Determine the matrix B_{FE}.

(e) Stability of the Forward Euler discretization

Using the eigenvalues of A, determine the condition on h under which the Forward Euler method is stable. Briefly explain how this condition relates to the stability analysis for scalar ODEs.

Solution 6

(a) Matrix form

The system can be written as,

$$\mathbf{u}'(t) = A\,\mathbf{u}(t), \qquad A = \begin{pmatrix} -\alpha & \beta \\ -\beta & -\alpha \end{pmatrix}.$$

(b) Eigenvalues and qualitative behavior

The eigenvalues of A satisfy,

$$\det(A - \lambda I) = \begin{vmatrix} -\alpha - \lambda & \beta \\ -\beta & -\alpha - \lambda \end{vmatrix} = (\alpha + \lambda)^2 + \beta^2 = 0.$$

Thus,

$$\boxed{\lambda_{1,2} = -\alpha \pm i\beta.}$$

Since the real part is negative, the solution decays in time, while the imaginary part induces oscillations. Hence, the system exhibits *damped oscillatory behavior* (spiral decay toward the origin).

(c) Exact solution structure

Define,

$$r(t)^2 = x(t)^2 + y(t)^2.$$

Differentiate,

$$\frac{d}{dt} r(t)^2 = 2xx' + 2yy'.$$

Substitute the system,

$$\begin{aligned} \frac{d}{dt} r(t)^2 &= 2x(-\alpha x + \beta y) + 2y(-\beta x - \alpha y) \\ &= -2\alpha(x^2 + y^2) + 2\beta(xy - xy) \\ &= -2\alpha r(t)^2. \end{aligned}$$

Therefore,

$$\frac{d}{dt} r(t)^2 = -2\alpha r(t)^2,$$

which is a scalar exponential decay equation. Solving,

$$\boxed{r(t)^2 = r(0)^2 e^{-2\alpha t}, \qquad r(t) = r(0) e^{-\alpha t}.}$$

Thus the magnitude of the solution decays exponentially, even though the components oscillate.

(d)

Forward Euler applied to $\mathbf{u}' = A\mathbf{u}$ gives,

$$\mathbf{u}_{n+1} = \mathbf{u}_n + hA\mathbf{u}_n = (I + hA)\mathbf{u}_n.$$

Hence,

$$\boxed{B_{\mathrm{FE}} = \begin{pmatrix} 1 - h\alpha & h\beta \\ -h\beta & 1 - h\alpha \end{pmatrix}.}$$

(e)

The eigenvalues of B_{FE} are,

$$\mu_{1,2} = 1 + h\lambda_{1,2} = 1 - h\alpha \pm ih\beta.$$

Forward Euler is stable if $|\mu_{1,2}| \leq 1$, i.e.,

$$(1 - h\alpha)^2 + (h\beta)^2 \leq 1.$$

Expanding,

$$1 - 2h\alpha + h^2(\alpha^2 + \beta^2) \leq 1 \quad \Longleftrightarrow \quad h\big(\alpha^2 + \beta^2\big) \leq 2\alpha.$$

Thus the stability condition is,

$$\boxed{0 < h \leq \frac{2\alpha}{\alpha^2 + \beta^2}.}$$

This mirrors the scalar case: stability is controlled by the real part of the eigenvalues, but oscillatory components (β) make the step-size restriction more severe.